Quiz 2 Solutions (5 Points)

Given the following instructions.

LDI R16, 0X35

LDI R17, 0X79

ADD R16, R17

```
          1  11          1
C=0    0011 0101
      +  0111 1001
        1010   1110
```

H = 1

N = 1

V = 1 (+ plus + = -)

S = N xor V = 1 xor 1 = 0

Z = 0 (non-zero result)

What is the value of the carry bit (C)? C = 0

What is the value of the negative bit (N)? N = 1 (most significant bit value)

What is the value of the overflow bit (V)? V = 1

What is the value of the carry bit (S)? S = 0

What is the value of the zero bit (Z)? Z = 0

What is the binary value of the sum in R16? 1010 1110